Contents

1 Appendix F: Plasmonic Nano-Geometry Sweep

1.1 Objective

Comprehensive plasmonic nanostructure analysis: frequency-dependent permittivity (Drude), Mie scattering, coupled-dipole near-field interactions, geometry optimization, and field-enhancement mapping for receptor-scale enhancement.

1.2 Methods (src)

- `src/case_studies/plasmonic_geometry.py`
  - `drude_model_permittivity(frequency_hz, metal_type)` — material permittivity model
  - `mie_scattering_sphere(radius_m, wavelength_m, eps_particle, eps_medium)` — Mie solutions
  - `coupled_dipoles_near_field(positions, polarizabilities, wavelength)` — multi-particle interactions
  - `optimize_plasmonic_geometry(wavelength_range, constraints)` — geometry optimization
  - `field_distribution_near_particle(particle_params, grid_points)` — near-field maps
  - `sweep_plasmonic_quality(radii_m, metal_eps, medium_eps)` — parameter sweeps for Q-factor analysis

1.3 Script and outputs

- Script: `scripts/generate_plasmonic_geometry_sweep.py`
- Data: `output/data/plasmonic_geometry_comprehensive.npz`
- Figure: `figures/plasmonic_geometry_comprehensive_analysis.png`

1.4 Figure

1.5 Equation references

– Resonance/wavelength: see main text and the Mathematical Appendix sec:mathematical$_a$*ppendix*.

1.6 Reproducibility

- Run: `python3 scripts/generate_plasmonic_geometry_sweep.py`
- Artifacts saved to `output/data/` and `figures/`.

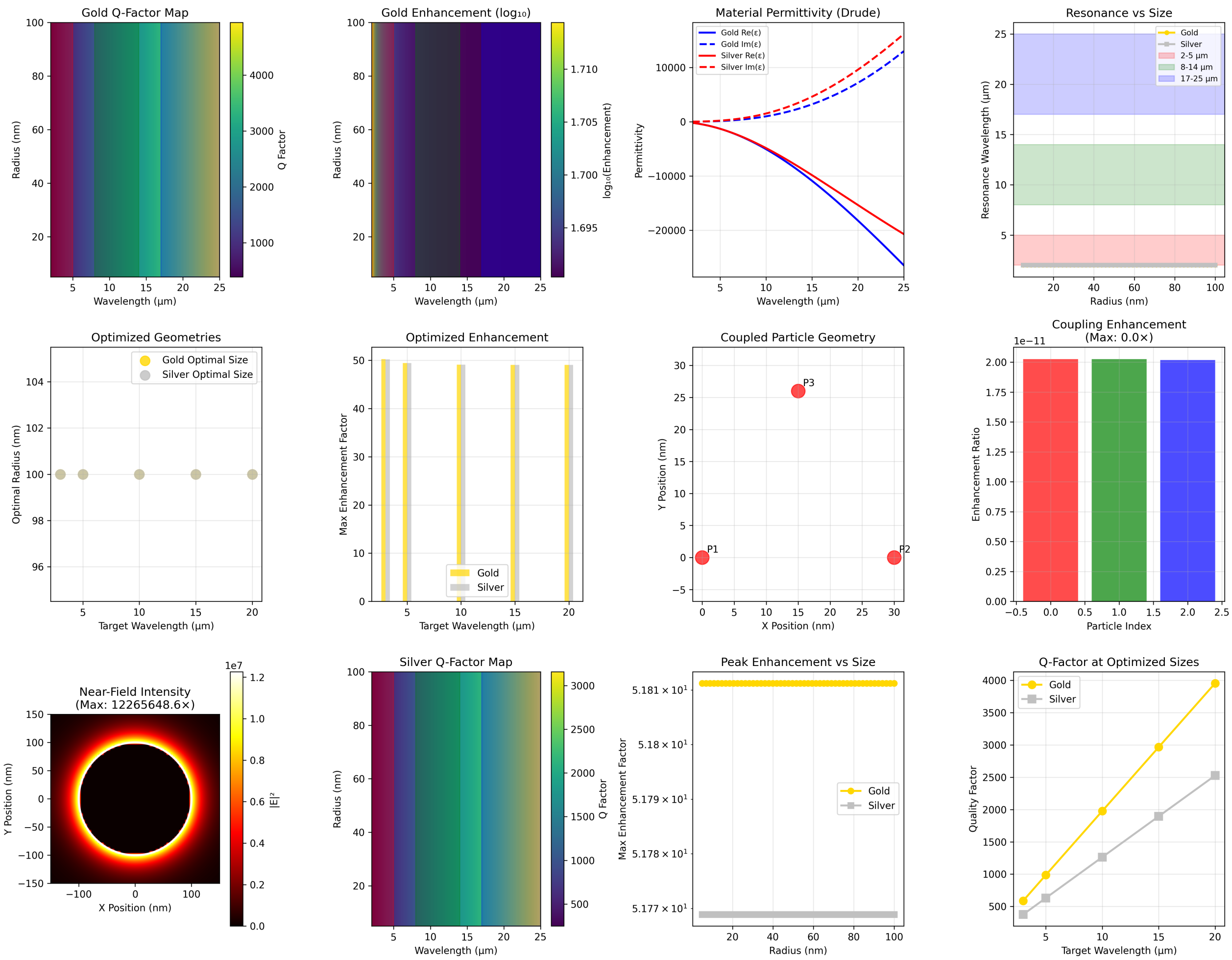

Figure 1: *Plasmonic geometry sweep produced deterministically by* $`scripts/generate_plasmonic_geometry_sweep.py`using`src/case_studies/plasmonic_geometry.py`.PanelsshowDrudepermitti$

- Deterministic radii grid and material parameters via `src/config.set_random_seed(42)`.

1.7 Cross-references

- Methods: sec:methodology
- Symbols: sec:symbols$_glossaryMathappendix : sec : mathematical_appendix$

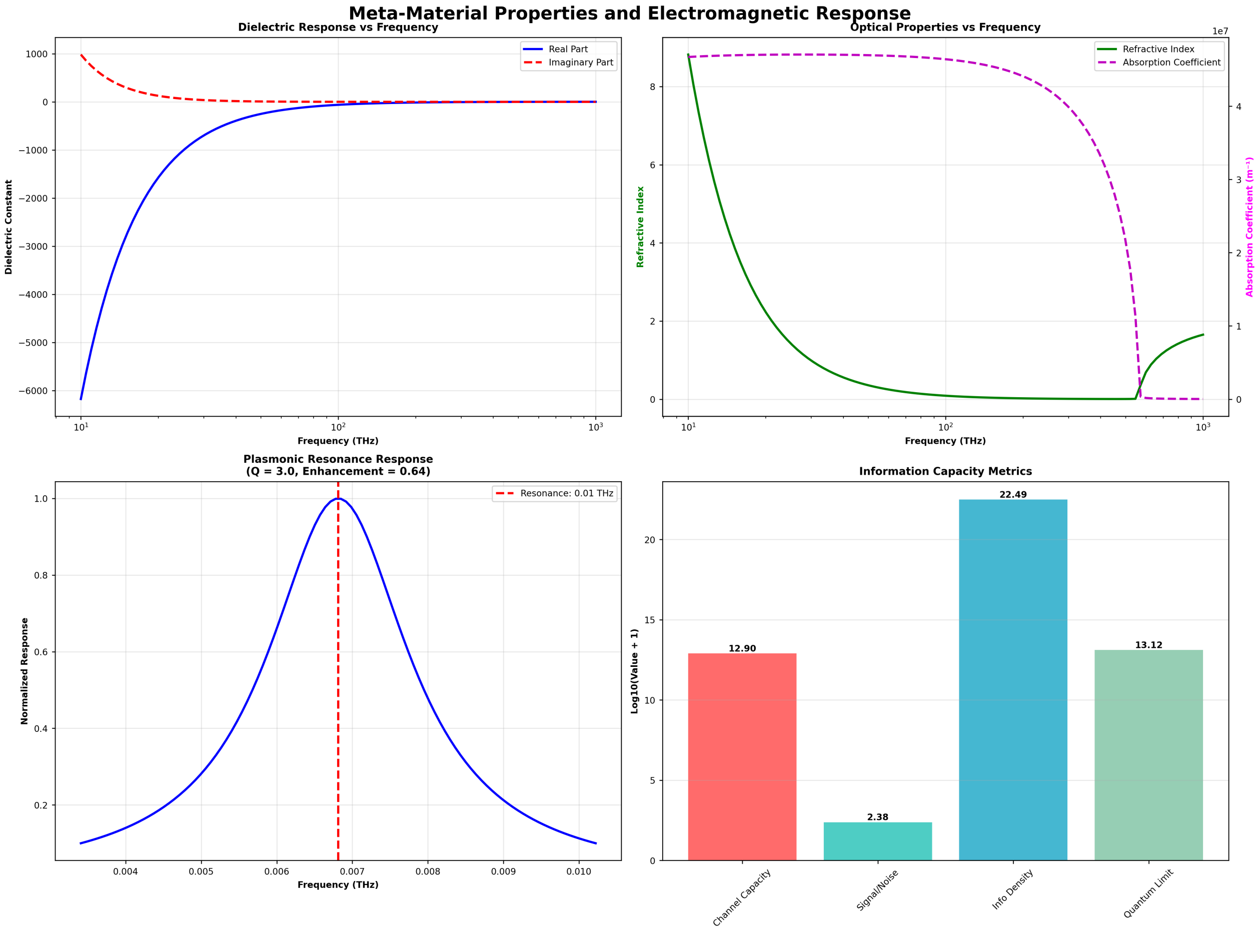

Figure 2: *Integrated metamaterial properties (dielectric and plasmonic summaries) produced by* ‘$scripts/generate_integrated_analysis.py$‘$.Panelssummarizedielectricresponse, plasmonicresonance, andinformation$-$ca$